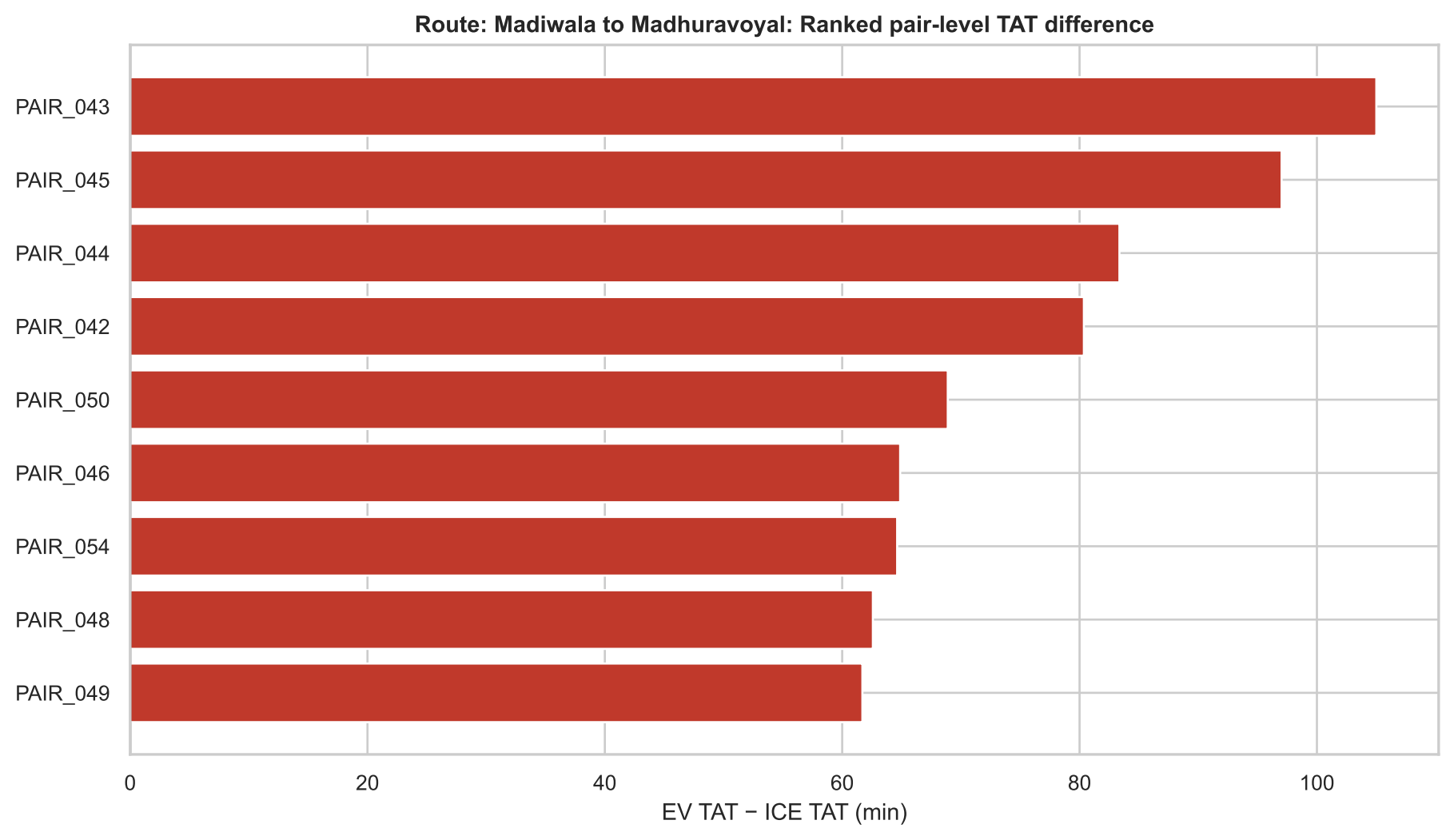
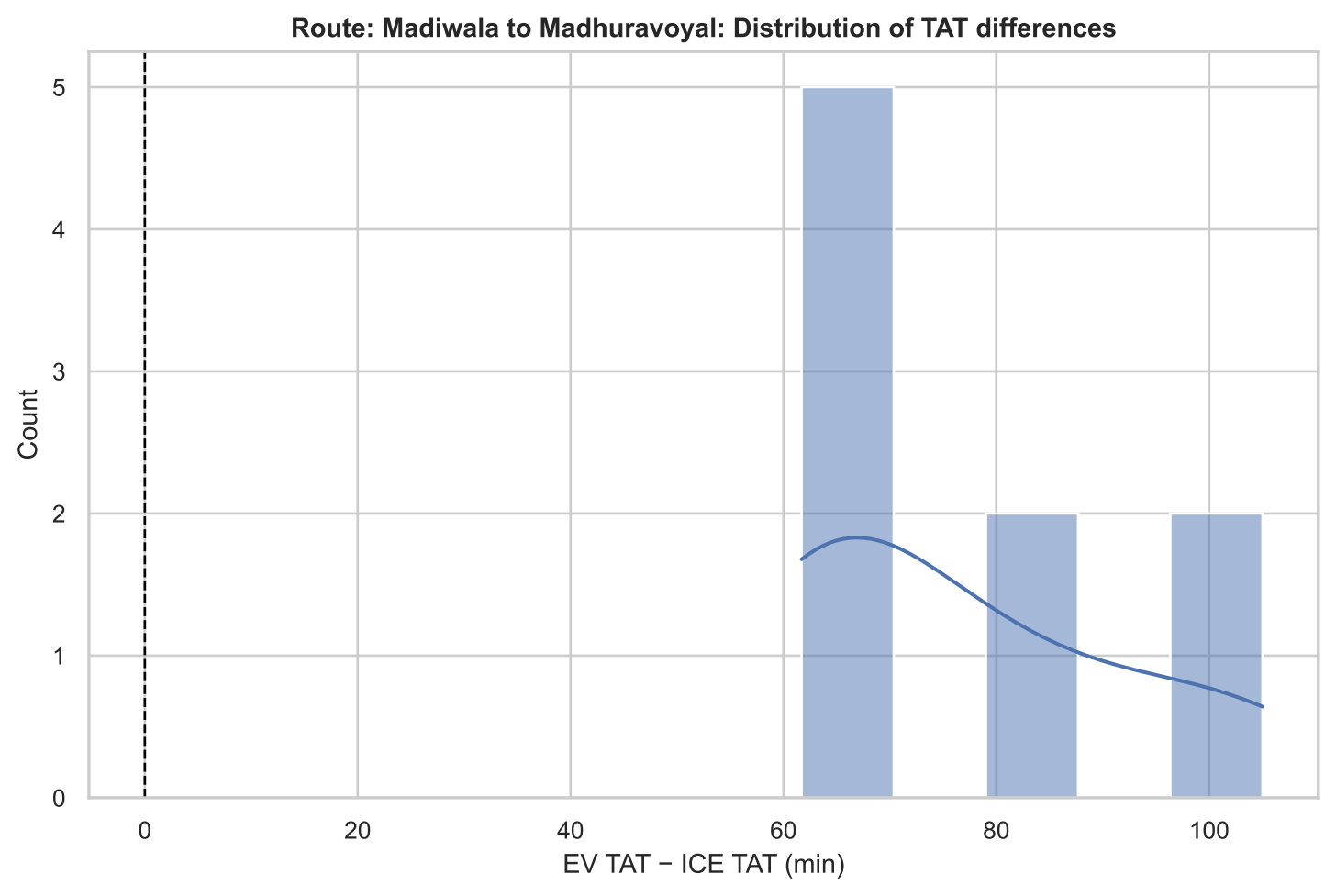
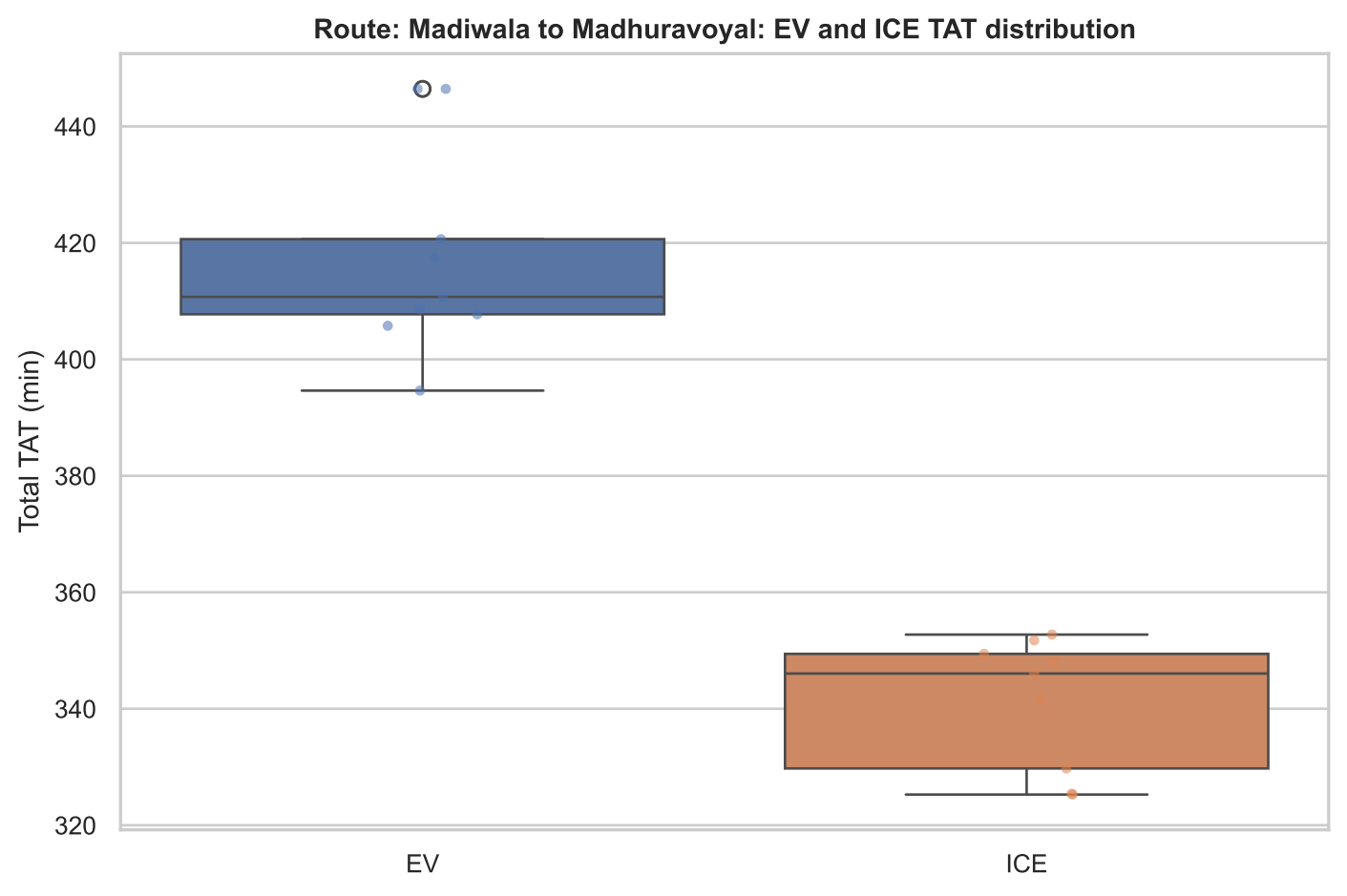
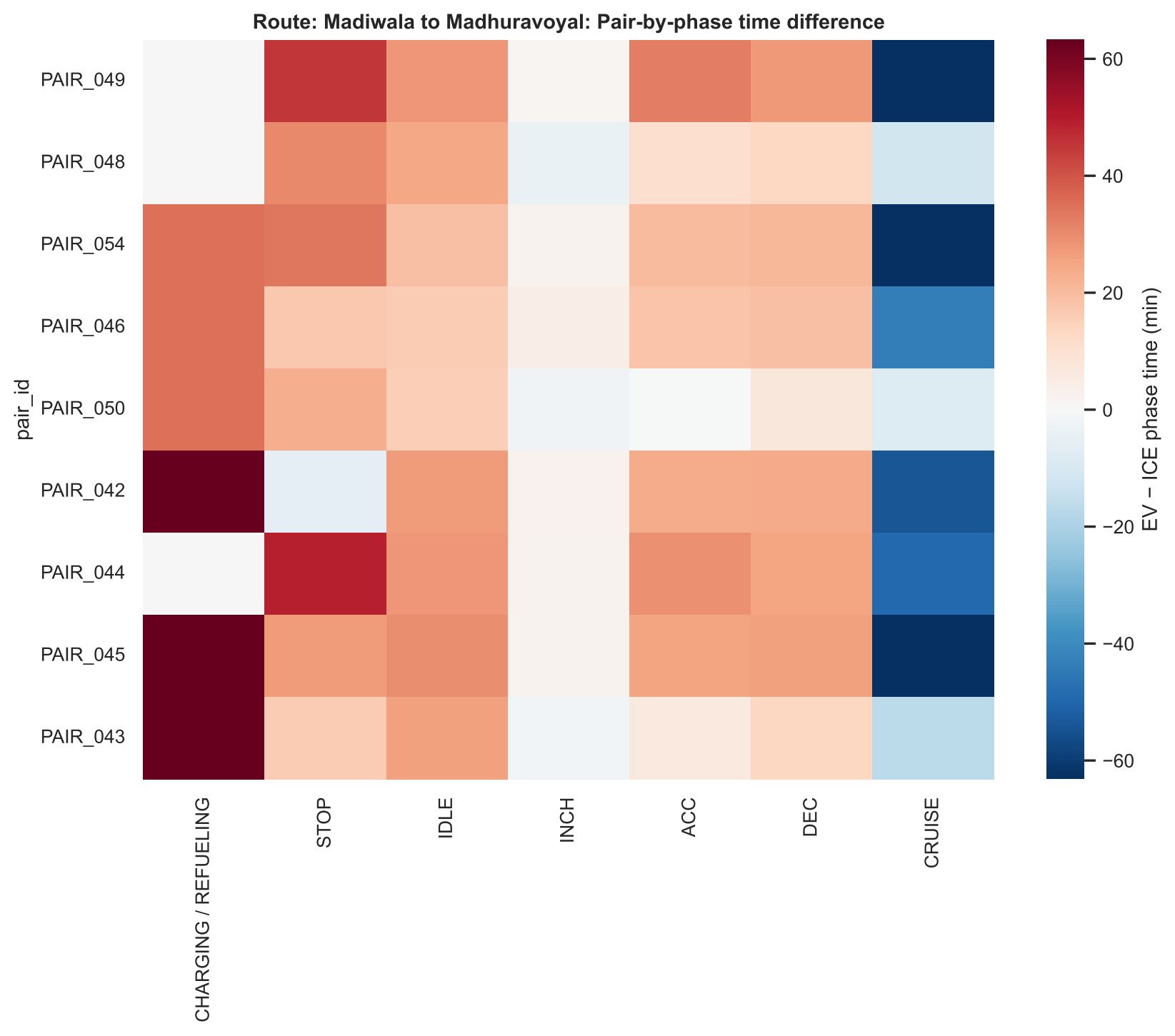
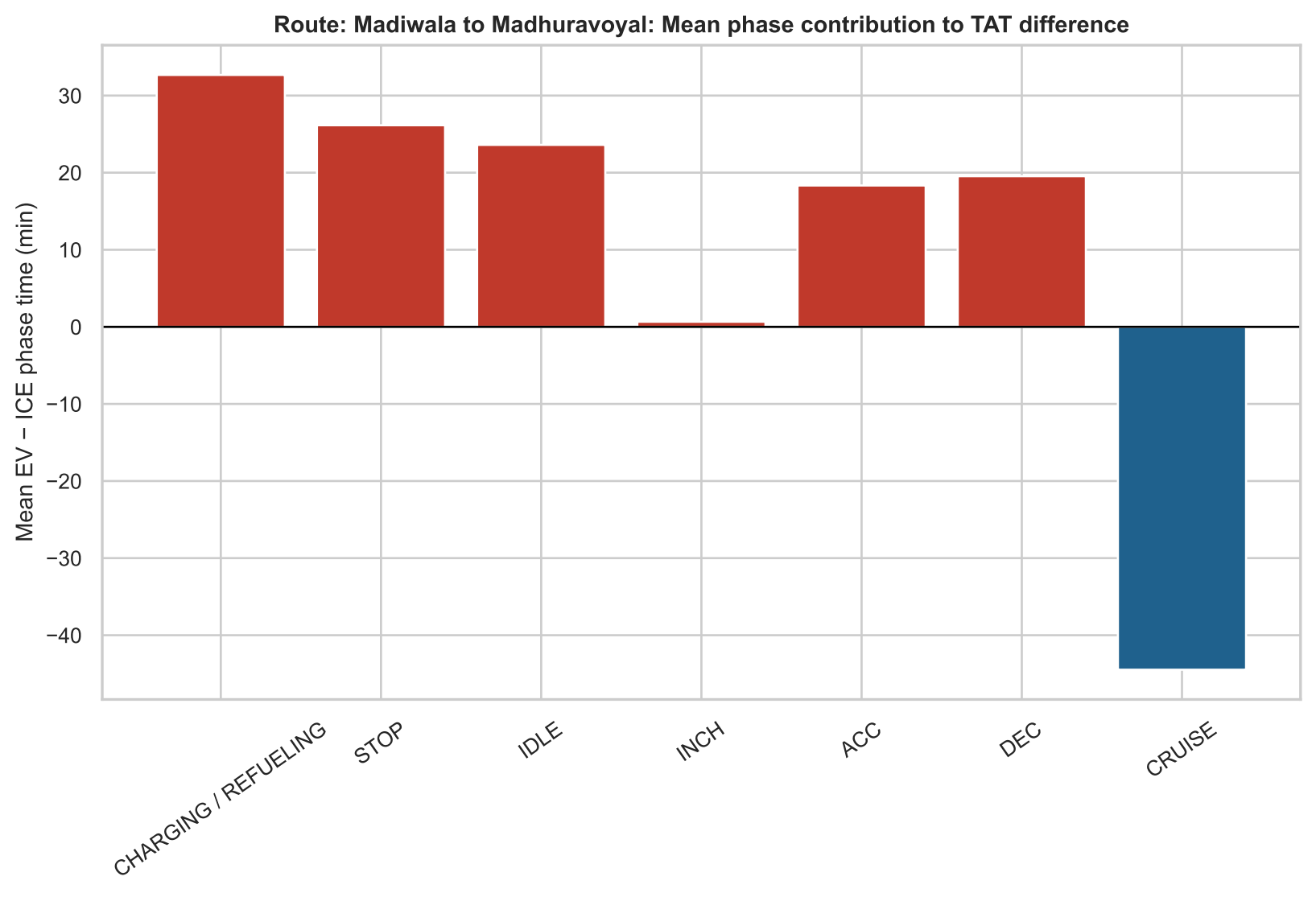
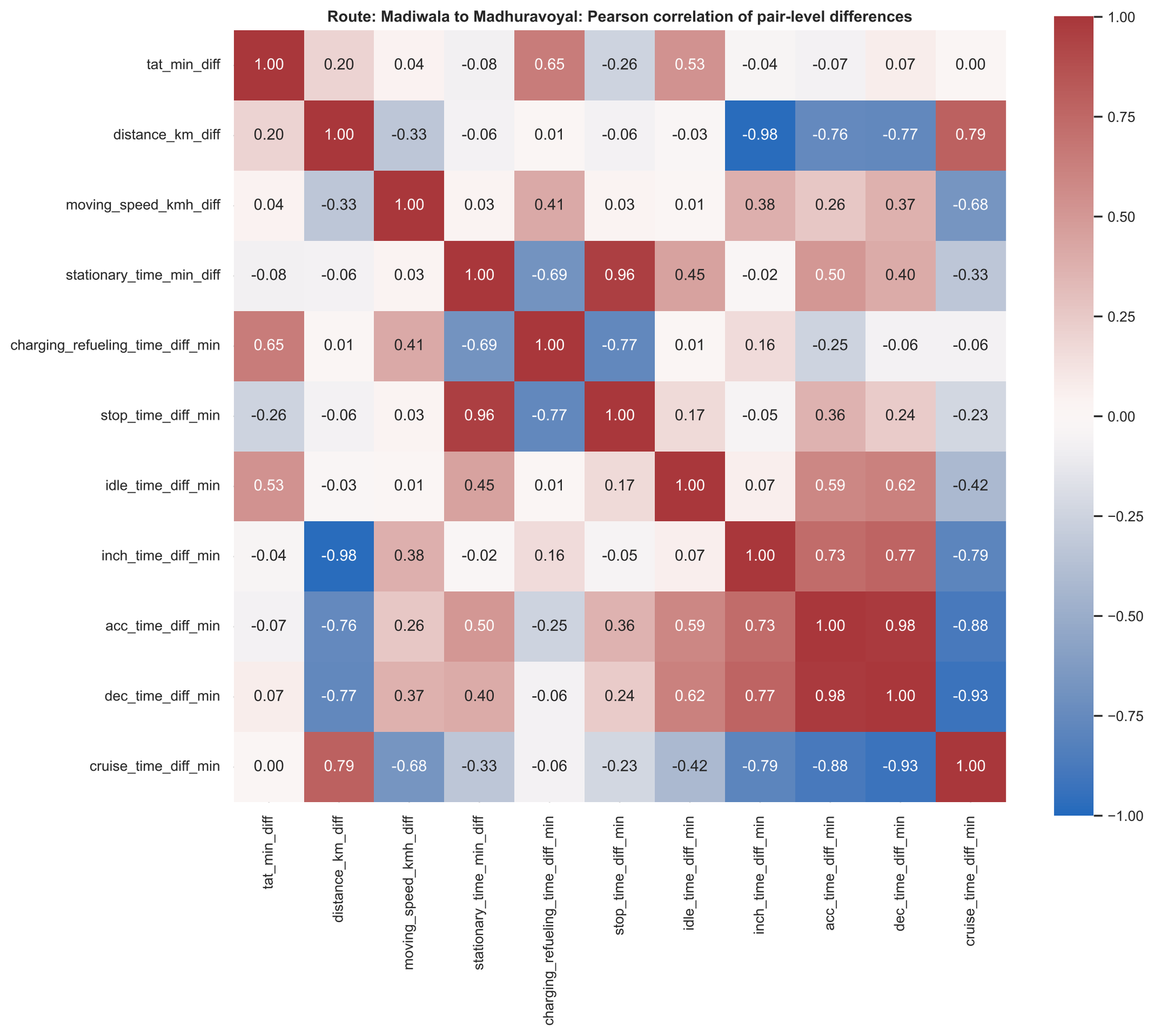

Route: Madiwala to Madhuravoyal: Ranked pair-level TAT difference
PAIR_043
PAIR_045
PAIR_044
PAIR_042
PAIR_050
PAIR_046
PAIR_054
PAIR_048
PAIR_049
0
20
40
60
80
100
EV TAT − ICE TAT (min)

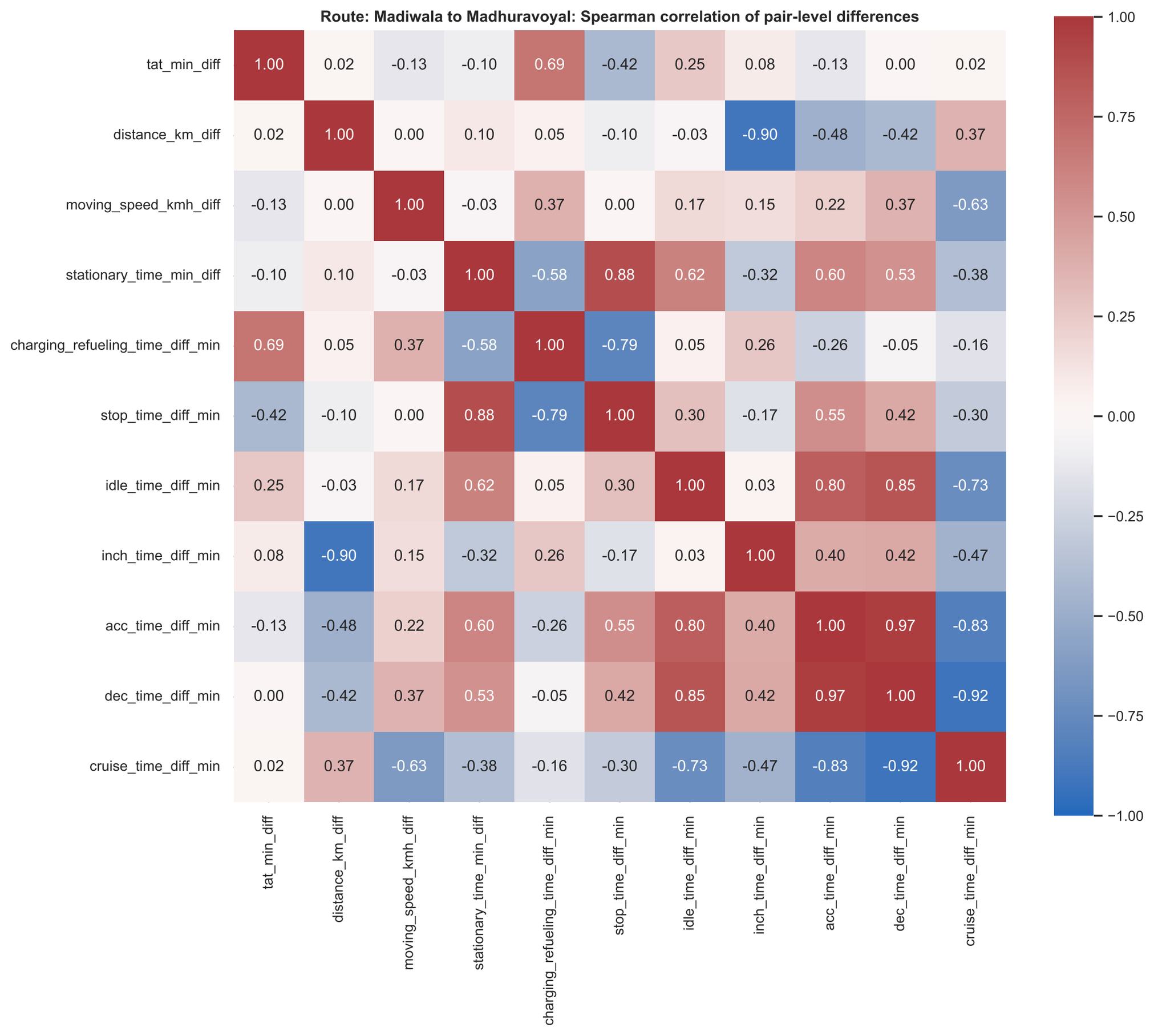

Route: Madiwala to Madhuravoyal: Distribution of TAT differences
Count
0
1
2
3
4
5
0
20
40
60
80
100
EV TAT − ICE TAT (min)

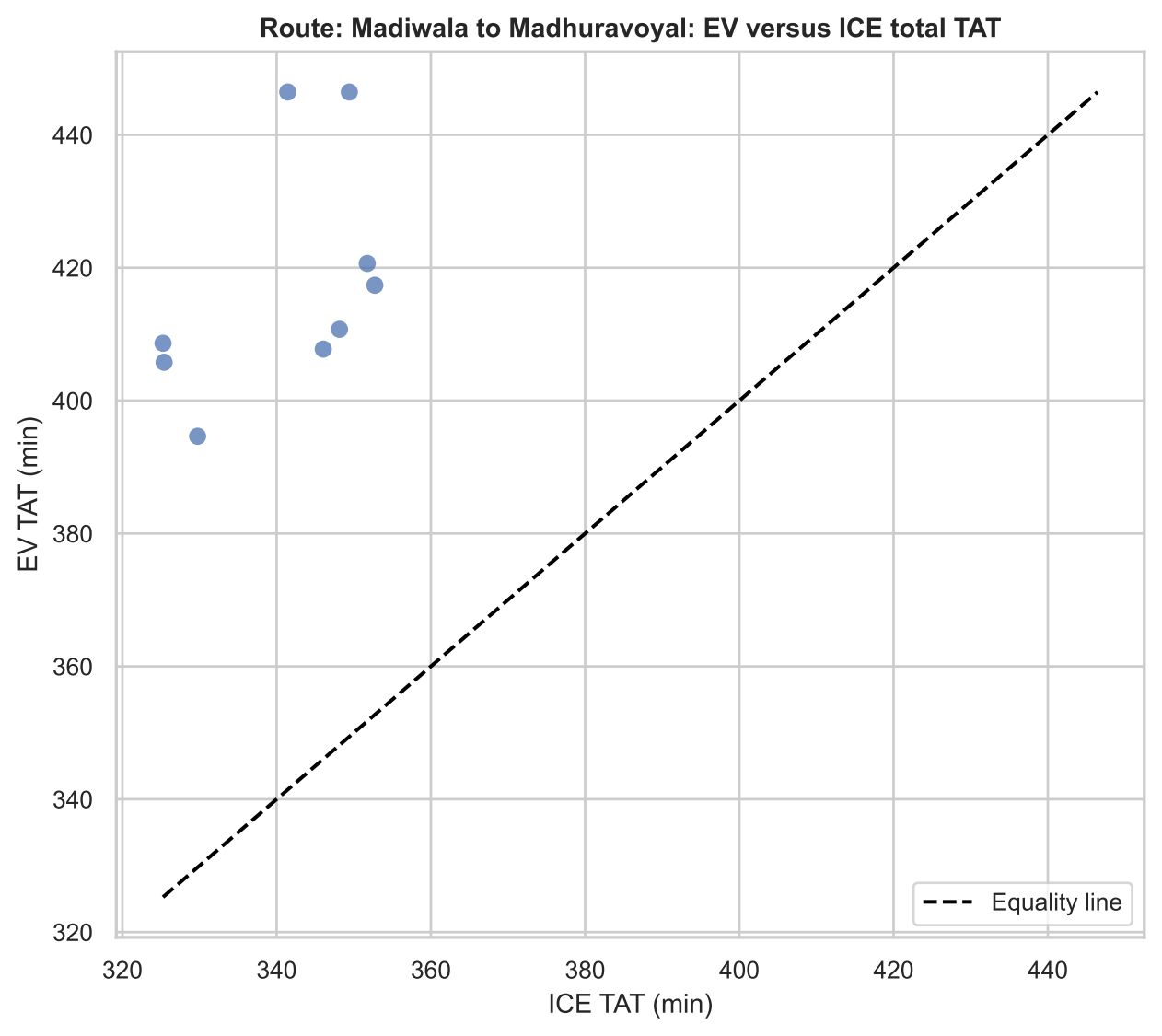

Route: Madiwala to Madhuravoyal: EV and ICE TAT distribution
Total TAT (min)
440
420
400
380
360
340
320
EV
ICE

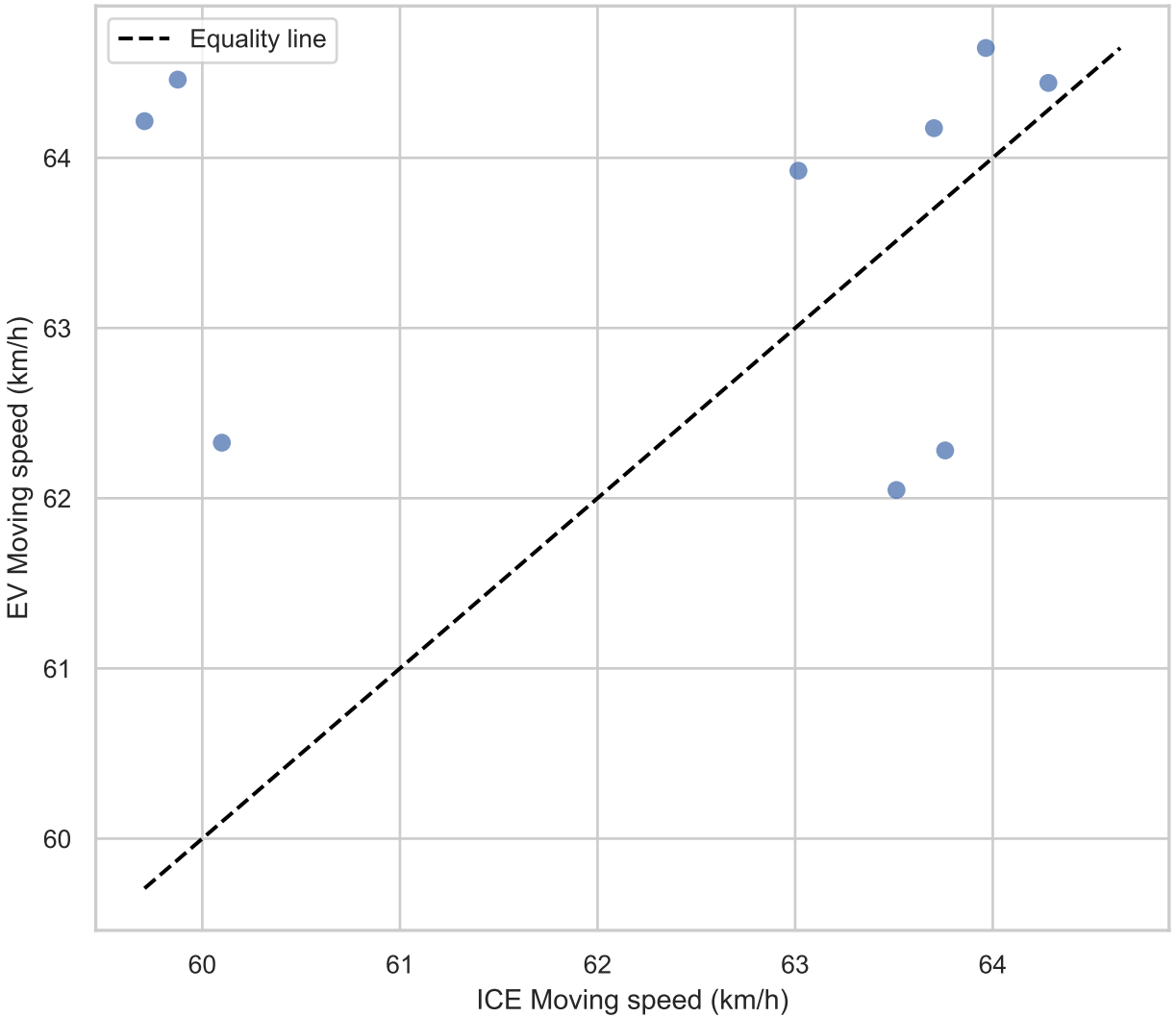

Route: Madiwala to Madhuravoyal: Pair-by-phase time difference
pair_id
PAIR_049
PAIR_048
PAIR_054
PAIR_046
PAIR_050
PAIR_042
PAIR_044
PAIR_045
PAIR_043
CHARGING / REFUELING
STOP
IDLE
INCH
ACC
DEC
CRUISE
EV − ICE phase time (min)
60
40
20
0
−20
−40
−60

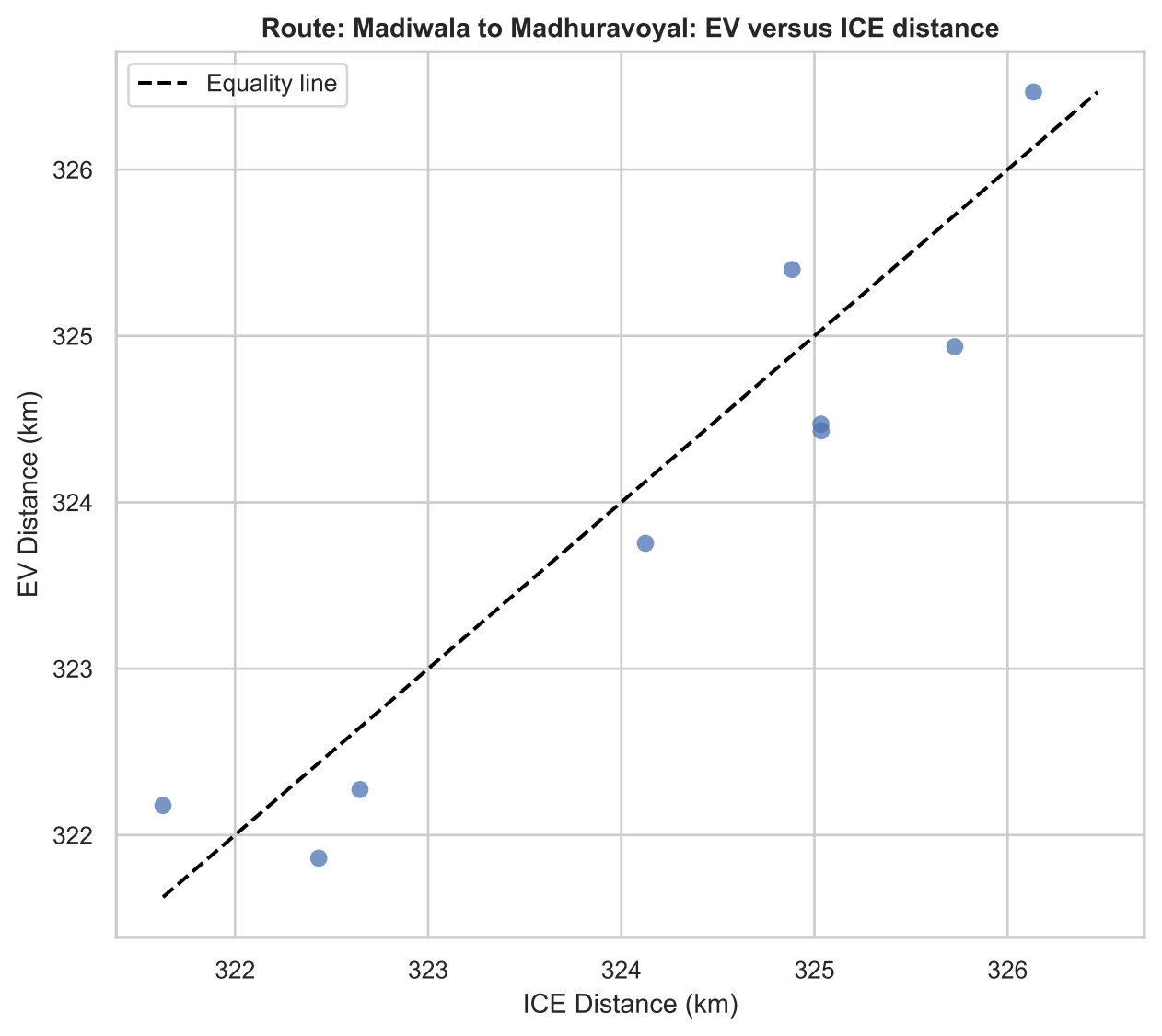

Route: Madiwala to Madhuravoyal: Mean phase contribution to TAT difference
Mean EV – ICE phase time (min)
30
20
10
0
−10
−20
−30
−40
CHARGING / REFUELING
STOP
IDLE
INCH
ACC
DEC
CRUISE

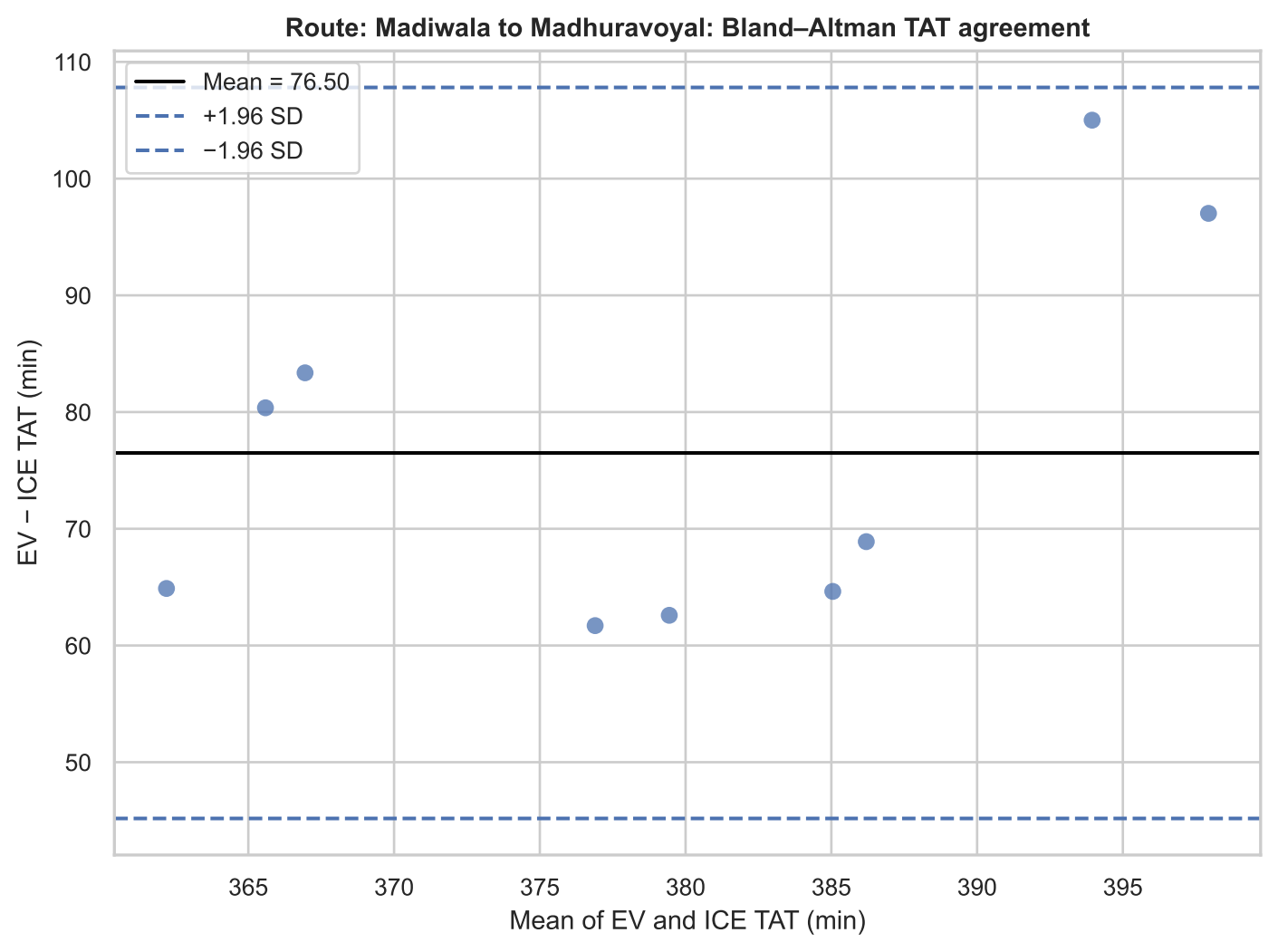

Route: Madiwala to Madhuravoyal: Pearson correlation of pair-level differences
tat_min_diff 1.00 0.20 0.04 -0.08 0.65 -0.26 0.53 -0.04 -0.07 0.07 0.00
distance_km_diff 0.20 1.00 -0.33 -0.06 0.01 -0.06 -0.03 -0.98 -0.76 -0.77 0.79
moving_speed_kmh_diff 0.04 -0.33 1.00 0.03 0.41 0.03 0.01 0.38 0.26 0.37 -0.68
stationary_time_min_diff -0.08 -0.06 0.03 1.00 -0.69 0.96 0.45 -0.02 0.50 0.40 -0.33
charging_refueling_time_diff_min 0.65 0.01 0.41 -0.69 1.00 -0.77 0.01 0.16 -0.25 -0.06 -0.06
stop_time_diff_min -0.26 -0.06 0.03 0.96 -0.77 1.00 0.17 -0.05 0.36 0.24 -0.23
idle_time_diff_min 0.53 -0.03 0.01 0.45 0.01 0.17 1.00 0.07 0.59 0.62 -0.42
inch_time_diff_min -0.04 -0.98 0.38 -0.02 0.16 -0.05 0.07 1.00 0.73 0.77 -0.79
acc_time_diff_min -0.07 -0.76 0.26 0.50 -0.25 0.36 0.59 0.73 1.00 0.98 -0.88
dec_time_diff_min 0.07 -0.77 0.37 0.40 -0.06 0.24 0.62 0.77 0.98 1.00 -0.93
cruise_time_diff_min 0.00 0.79 -0.68 -0.33 -0.06 -0.23 -0.42 -0.79 -0.88 -0.93 1.00
tat_min_diff
distance_km_diff
moving_speed_kmh_diff
stationary_time_min_diff
charging_refueling_time_diff_min
stop_time_diff_min
idle_time_diff_min
inch_time_diff_min
acc_time_diff_min
dec_time_diff_min
cruise_time_diff_min
1.00
0.75
0.50
0.25
0.00
−0.25
−0.50
−0.75
−1.00

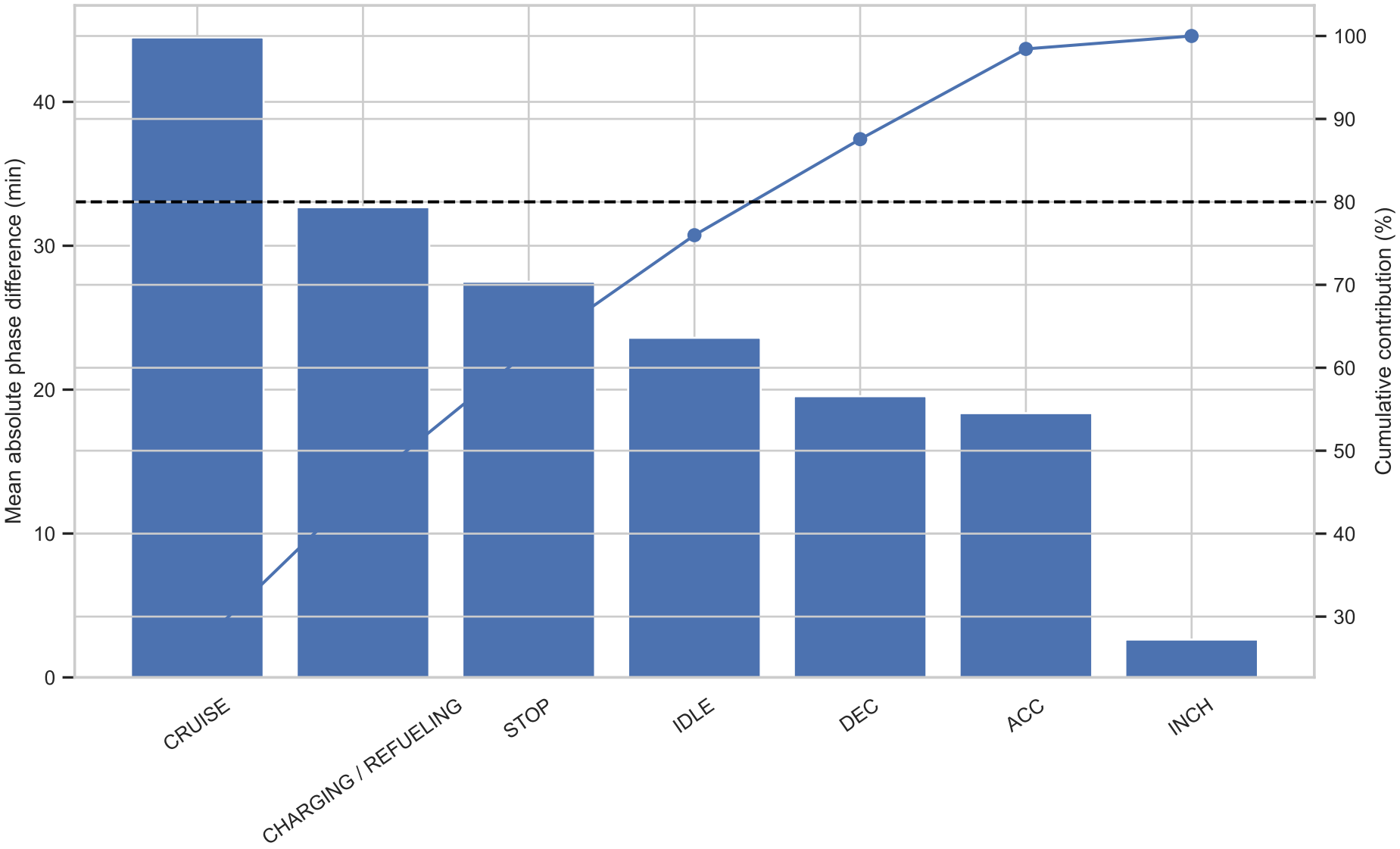

Route: Madiwala to Madhuravoyal: Spearman correlation of pair-level differences
tat_min_diff 1.00 0.02 -0.13 -0.10 0.69 -0.42 0.25 0.08 -0.13 0.00 0.02
distance_km_diff 0.02 1.00 0.00 0.10 0.05 -0.10 -0.03 -0.90 -0.48 -0.42 0.37
moving_speed_kmh_diff -0.13 0.00 1.00 -0.03 0.37 0.00 0.17 0.15 0.22 0.37 -0.63
stationary_time_min_diff -0.10 0.10 -0.03 1.00 -0.58 0.88 0.62 -0.32 0.60 0.53 -0.38
charging_refueling_time_diff_min 0.69 0.05 0.37 -0.58 1.00 -0.79 0.05 0.26 -0.26 -0.05 -0.16
stop_time_diff_min -0.42 -0.10 0.00 0.88 -0.79 1.00 0.30 -0.17 0.55 0.42 -0.30
idle_time_diff_min 0.25 -0.03 0.17 0.62 0.05 0.30 1.00 0.03 0.80 0.85 -0.73
inch_time_diff_min 0.08 -0.90 0.15 -0.32 0.26 -0.17 0.03 1.00 0.40 0.42 -0.47
acc_time_diff_min -0.13 -0.48 0.22 0.60 -0.26 0.55 0.80 0.40 1.00 0.97 -0.83
dec_time_diff_min 0.00 -0.42 0.37 0.53 -0.05 0.42 0.85 0.42 0.97 1.00 -0.92
cruise_time_diff_min 0.02 0.37 -0.63 -0.38 -0.16 -0.30 -0.73 -0.47 -0.83 -0.92 1.00
tat_min_diff
distance_km_diff
moving_speed_kmh_diff
stationary_time_min_diff
charging_refueling_time_diff_min
stop_time_diff_min
idle_time_diff_min
inch_time_diff_min
acc_time_diff_min
dec_time_diff_min
cruise_time_diff_min
1.00
0.75
0.50
0.25
0.00
−0.25
−0.50
−0.75
−1.00

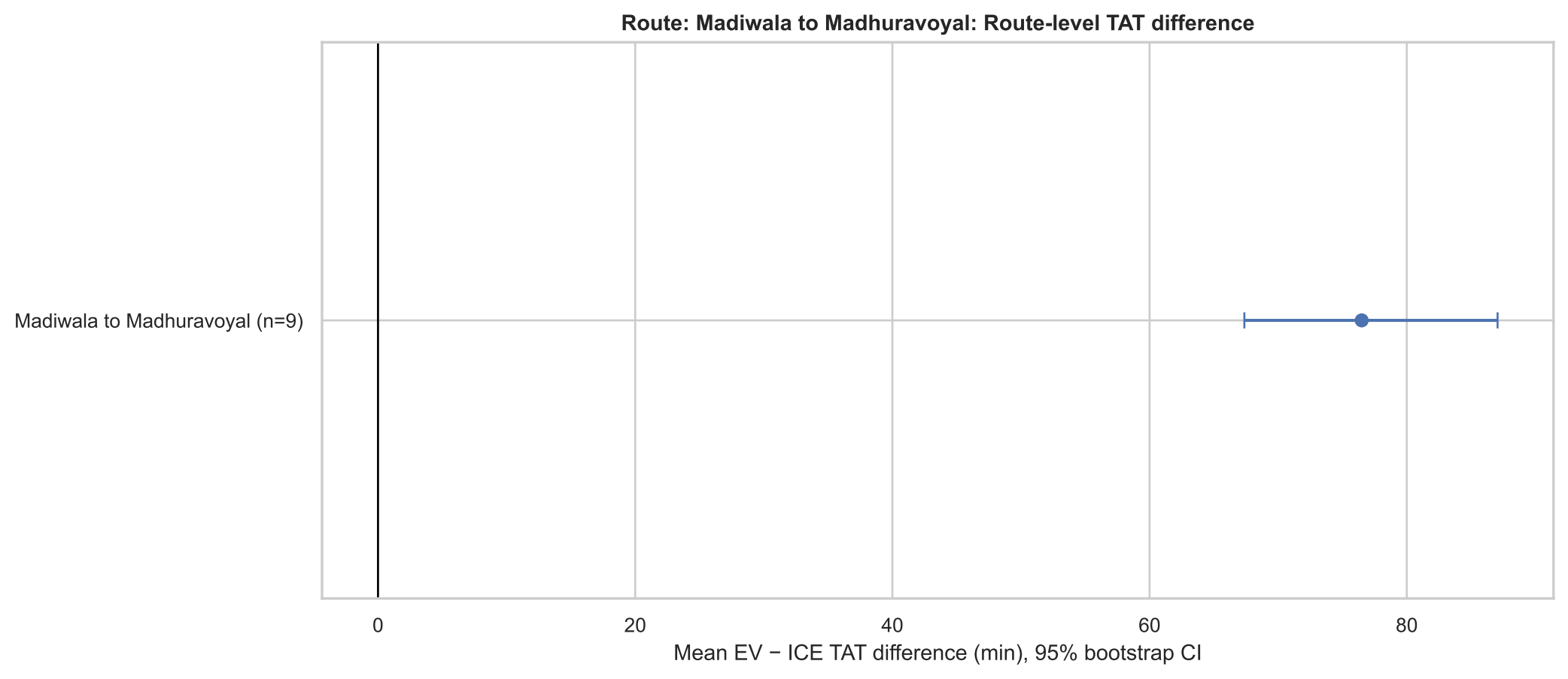

Route: Madiwala to Madhuravoyal: EV versus ICE total TAT
EV TAT (min)
ICE TAT (min)
320
340
360
380
400
420
440
Equality line

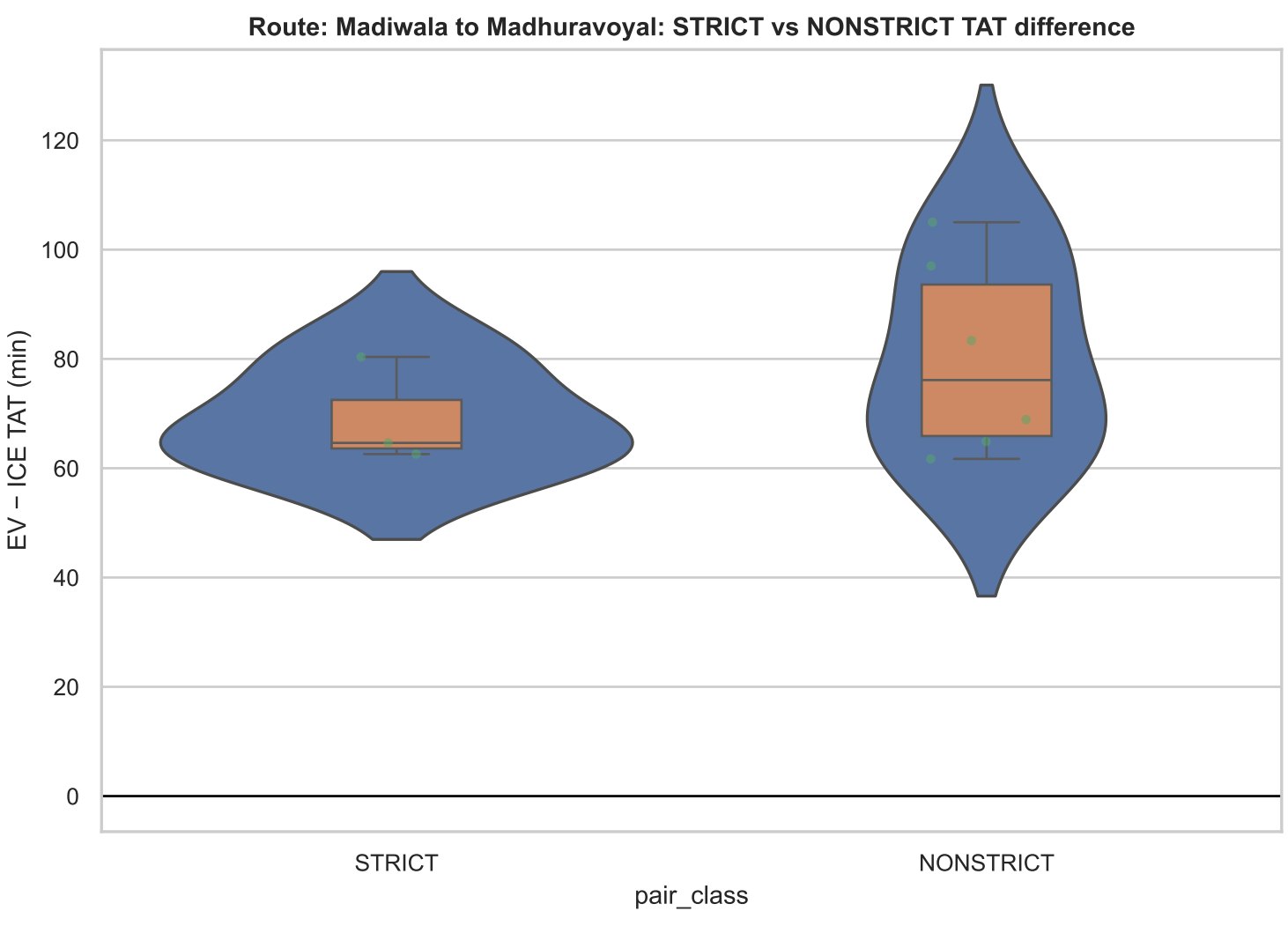

Route: Madiwala to Madhuravoyal: EV versus ICE moving speed
Equality line
EV Moving speed (km/h)
ICE Moving speed (km/h)
60
61
62
63
64

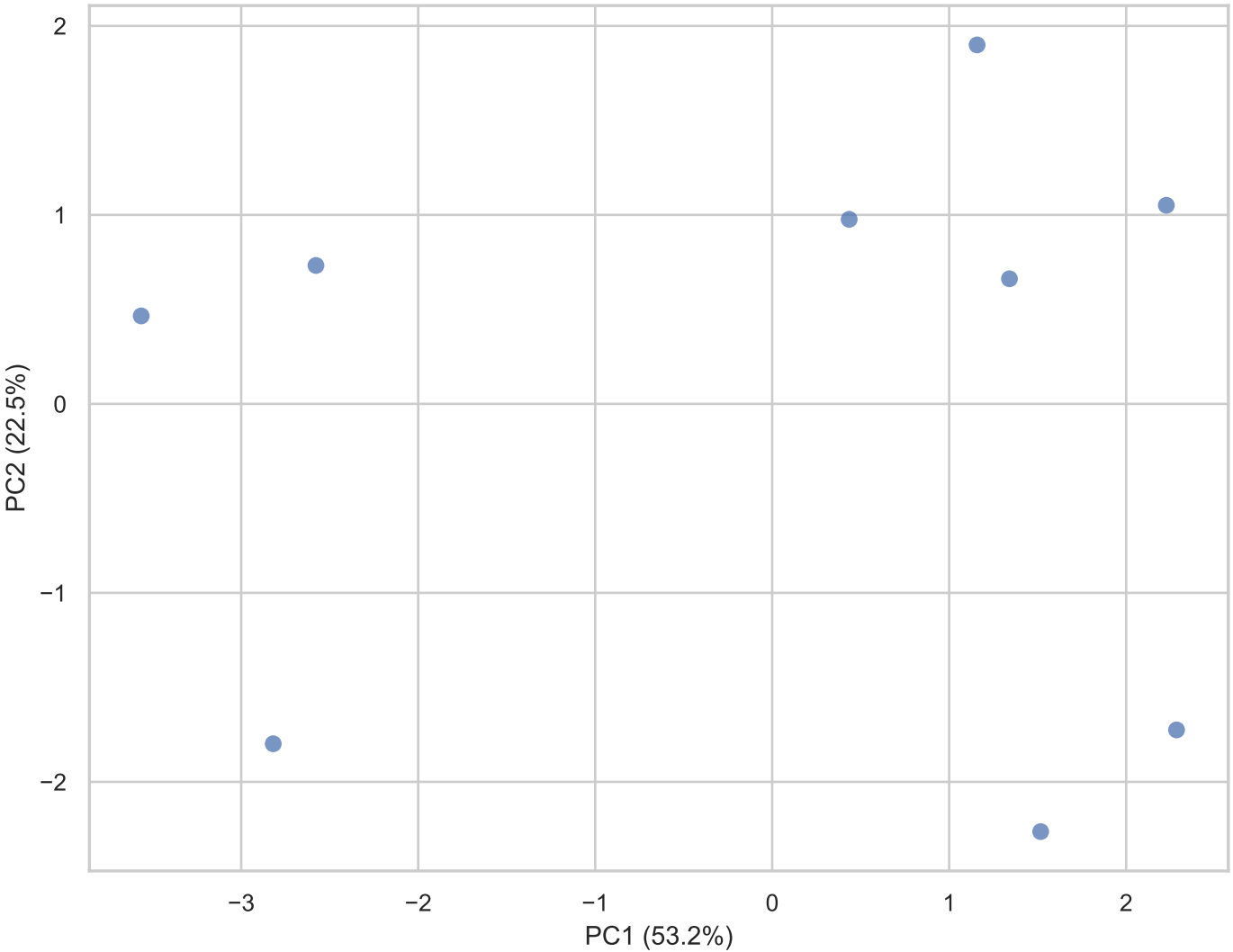

Route: Madiwala to Madhuravoyal: EV versus ICE distance
Equality line
EV Distance (km)
ICE Distance (km)
322
323
324
325
326

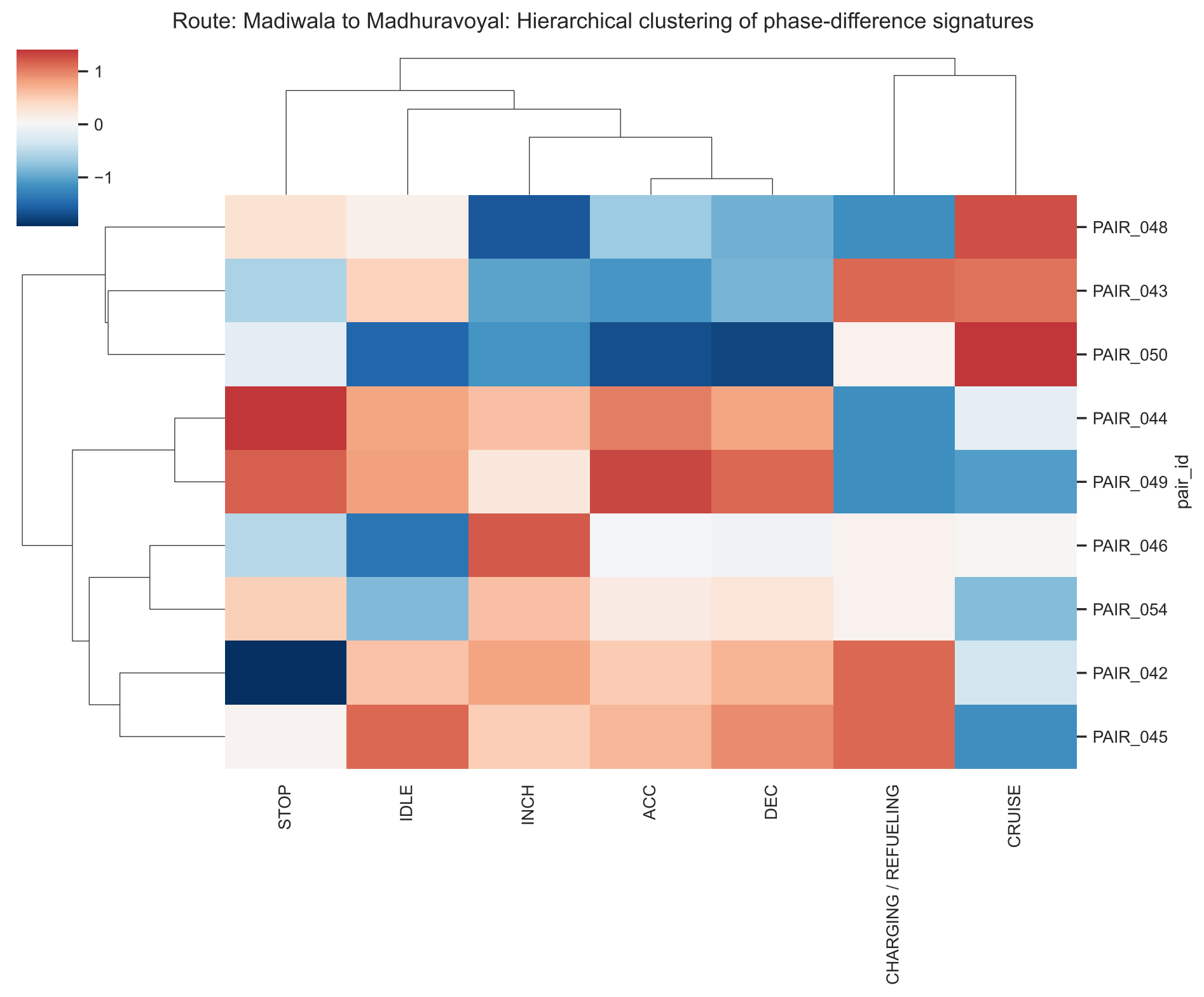

Route: Madiwala to Madhuravoyal: Bland–Altman TAT agreement
Mean = 76.50
+1.96 SD
−1.96 SD
110
100
90
80
70
60
50
EV – ICE TAT (min)
365
370
375
380
385
390
395
Mean of EV and ICE TAT (min)


Route: Madiwala to Madhuravoyal: Pareto analysis of phase effects
Mean absolute phase difference (min)
Cumulative contribution (%)
0
10
20
30
40
30
40
50
60
70
80
90
100
CRUISE
CHARGING / REFUELING
STOP
IDLE
DEC
ACC
INCH


Route: Madiwala to Madhuravoyal: Route-level TAT difference
Madiwala to Madhuravoyal (n=9)
0
20
40
60
80
Mean EV − ICE TAT difference (min), 95% bootstrap CI


Route: Madiwala to Madhuravoyal: STRICT vs NONSTRICT TAT difference
EV – ICE TAT (min)
120
100
80
60
40
20
0
STRICT
NONSTRICT
pair_class


Route: Madiwala to Madhuravoyal: PCA of EV−ICE operational differences
PC2 (22.5%)
PC1 (53.2%)
2
1
0
−1
−2
−3
−2
−1
0
1
2


Route: Madiwala to Madhuravoyal: Hierarchical clustering of phase-difference signatures
1
0
−1
PAIR_048
PAIR_043
PAIR_050
PAIR_044
PAIR_049
PAIR_046
PAIR_054
PAIR_042
PAIR_045
pair_id
STOP
IDLE
INCH
ACC
DEC
CHARGING / REFUELING
CRUISE